Utkarsh Bharadwaj

Email | LinkedIn | GitHub | Portfolio Website

About Me

Statistics student at the Indian Statistical Institute, Delhi. Interested in applications of probability theory and statistical inference.

Education

Indian Statistical Institute **Delhi, India**

Bachelor of Statistical Data Science (BSDS) *Aug 2025 – Present*

- Score: **70.4%** (1st Semester); **84%** (2nd Semester)
- **Coursework**: Probability Theory, Analysis, Statistical Inference, Optimization, Linear Models, Graph Theory

Technical Skills

Languages: Python, R, LaTeX

Libraries & Frameworks: NumPy, ggplot2, lattice

Statistical Methods: Poisson Processes, QP/Convex Optimization, Regression Analysis, Hypothesis Testing

Tools & Environment: Linux (Debian), VS Code, RStudio, GitHub

Projects

The Measure of Normality *Oct 2026*

Normal Numbers · Probability Theory · Lebesgue Measure

- Revisited Borel's (1909), Hausdorff's (1914), and Weitz's (2021) proofs that almost all real numbers are normal, alongside the Lebesgue measure and extensions of the real line.

Neyman-Pearson Lab: Hypothesis Testing Tool *July 2026*

Hypothesis Testing · MP Test

- Built an interactive web app deriving MP (Neyman-Pearson), UMP (Karlin-Rubin), and UMPU level-α tests for nine distribution families.

CipherInk: Forensic Stylometry Platform *June 2026*

Bayesian Inference · Statistical NLP

- Developed a full-stack forensic stylometry web app for authorship attribution using Poisson log-likelihood and Bayesian softmax on 70 filler words.

Beyond the Deficit: India's Trade Dynamics *May 2026*

R · Time-Series Analysis · Macroeconomics

- Conducted a 35-year time-series analysis (1990–2025) of India's trade trends, evaluating currency and energy impacts using inflation-adjusted values.

Convex Optimization for LPG Allocation *Apr 2026 – May 2026*

QP · Optimization Theory

- Formulated a strictly convex QP model allocating India's constrained LPG supply across sectors, minimizing weighted squared shortfalls using historical consumption data.

Certifications & Workshops

Indian Institute of Technology Ropar **Ropar, India**
NS2ML-DE 2026 Workshop *Jul 6 – 10, 2026*

- Covered numerical PDE/ODE solvers and Physics-Informed Neural Networks with MATLAB/Python labs.

Achievements

ISI Merit List: AIR **31** for BSDS

JEE Main 2025: **99.37 percentile** in Mathematics (top 0.6%)

Hackathons *(Team Diazonium)*: Vihaan 9.0 Finalist (DTU); NIT Delhi UpVision Top 10 among 150+ teams